

1. Administrative & Study Identification

Full Study Title: A Phase 3, Randomized, Double-Blind, Placebo-Controlled, Parallel-Group Study to Evaluate the Efficacy and Safety of KarXT + KarX-EC for the Treatment of Cognitive Impairment Associated With Mild to Moderate Alzheimer's Disease (MINDSET 1) **Short Title/Acronym:** MINDSET 1 **Protocol Number:** CN012-0051 **NCT Number:** NCT06976216 **Sponsor Name:** Bristol-Myers Squibb **Investigational Product:** KarXT (xanomeline/trospium chloride) + KarX-EC **Phase of Development:** Phase 3 **Medical Monitor:** Bristol-Myers Squibb Clinical Operations

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of KarXT + KarX-EC compared with placebo for the treatment of cognitive impairment and global functioning in participants with mild to moderate Alzheimer's Disease. **Secondary Objectives:** To evaluate the long-term safety, tolerability, and additional measures of daily functioning and neuropsychiatric symptoms associated with Alzheimer's disease.

2.2 Background & Rationale To address the need for novel mechanisms of action in Alzheimer's disease symptom management. Current standard-of-care treatments for Alzheimer's cognitive impairment either have limited efficacy or target amyloid plaques/tau tangles directly. KarXT + KarX-EC offers a unique approach by modulating the muscarinic and nicotinic acetylcholine receptors, which may improve communication between brain cells and improve thinking skills without the severe side-effect profile of traditional antipsychotics.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized, Parallel-group

3.2 Planned Enrollment & Duration Target N: 586 participants **Number of Sites:** Multinational (Including sites in the US, Canada, Europe, and Rest of World) **Actual Study Start:** 14-Jul-2025 **Estimated Primary Completion:** 11-Sep-2028 **Estimated Study Completion:** 23-Feb-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adults (male and female) aged 60–85 years. 2. Confirmed diagnosis of Alzheimer's disease (AD) at the mild (stage 4) or moderate (stage 5) dementia stages, defined by the NIA-AA core clinical criteria. 3. Mini-Mental State Examination (MMSE) score ranging from 12 through 22, inclusive, at screening. 4. Must have a designated caregiver/study partner who maintains adequate contact (around 10 hours per week or more) and is willing to attend all study visits.

4.2 Exclusion Criteria 1. History of schizophrenia, chronic psychosis, major depression, schizoaffective disorder, or bipolar disorder. 2. Previous exposure to KarXT or current treatment with disease-modifying anti-amyloid therapies for AD within the past 6 months prior to screening. 3. Moderate-severe renal impairment (eGFR < 50 mL/min), hepatic impairment, urinary retention, active biliary disease, or unstable hypertension/tachycardia.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: KarXT and KarX-EC administered orally at specified doses on specified days. (KarX-EC is an enteric-coated formulation designed for intestinal release to improve tolerability). **Route of Administration:** Oral **Duration of Treatment:** 24 weeks for the primary evaluation period, followed by long-term follow up (estimated study completion in February 2029).

5.2 Concomitant & Prohibited Therapy Permitted: Acetylcholine esterase inhibitors (AChEIs) and/or memantine, provided the participant has been on a stable dosage for at least 12 weeks prior to screening and will maintain it during the study. **Prohibited:** Disease-modifying anti-amyloid therapies within 6 months prior to screening.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, MMSE, Brain MRI, Caregiver qualification
Baseline	Day 0	Randomization, First Dose, Caregiver reporting initiation	Interim Periodic Safety Labs, Efficacy Assessment, Caregiver Input interviews End of Study (Primary) Week 24 Final ADAS-Cog11, CIBIC+ evaluations, Adverse Event review

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change from baseline in the Alzheimer's Disease Assessment Scale-Cognitive Subscale 11 (ADAS-Cog11) at Week 24, and the Clinician's Interview-Based Impression Plus Caregiver Input (CIBIC+) at Week 24. **Measurement Method:** Standardized Clinical and Caregiver Assessment Scales.

7.2 Secondary & Exploratory Endpoints - Long-term safety and tolerability (e.g., incidence of gastrointestinal events like dry mouth and constipation). - Additional functional scales measuring activities of daily living and global functioning.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via onsite visits and patient/caregiver reporting. Frequent monitoring for known muscarinic/nicotinic modulation side effects (e.g., urinary retention, hepatic impairment). **Serious Adverse Events (SAEs):** Standard protocol; reported typically within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee oversight for trial safety.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy endpoints; Safety Set for all AE/SAE evaluations. **Sample Size Justification:** N=586, powered appropriately to detect a statistically significant difference in ADAS-Cog11 and CIBIC+ score changes between the KarXT + KarX-EC and placebo arms at Week 24. **Handling of Missing Data:** Standard handling for pivotal Alzheimer's trials (e.g., MMRM - Mixed-Effect Model Repeat Treatment).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to ensure integrity of cognitive scale administration. **Audit Trail:** Secure eCRF locking and centralized rating reviews to minimize inter-rater variability on the ADAS-Cog11 and CIBIC+.

11. Study Identifiers & Governance

Primary ID: CN012-0051 **Registry Number:** NCT06976216 **EudraCT Number:** 2025-520746-30 **Other Identifier (UTN):** U1111-1317-4462 **Sponsor/Lead Organization:** Bristol-Myers Squibb **Study Title:**

MINDSET 1 **Official Regulatory Title:** A Phase 3, Randomized, Double-Blind, Placebo-Controlled, Parallel-Group Study to Evaluate the Efficacy and Safety of KarXT + KarX-EC for the Treatment of Cognitive Impairment Associated With Mild to Moderate Alzheimer's Disease (MINDSET 1)

12. Clinical Framework (Alzheimer's Disease Specific)

Primary Condition: Cognitive Impairment in Alzheimer's Disease
Diagnosis Threshold: Mild to moderate dementia (NIA-AA criteria) with MMSE score 12-22
Medical Methodology: Muscarinic and nicotinic acetylcholine receptor modulation
Optimization Target: Improvement in cognitive functioning (ADAS-Cog11) and global impression (CIBIC+)

13. Study Procedures & Methodology

Management Pathway: Standard Alzheimer's clinical trial pathway
Education Protocol (HCP): Specialized rater training for consistent ADAS-Cog11 and CIBIC+ administration
Education Protocol (Patient): Caregiver involvement required (≥ 10 hours/week) for overseeing medication compliance and reporting
Quality Audit Mechanism: Rater tracking and electronic clinical outcome assessments (eCOA) monitoring

14. Observation & Follow-Up Schedule

Total Duration: Primary evaluation at 24 weeks; study continuation through Q1 2029
Onsite Visit Cadence: Regular assessment visits including Screening, Baseline, and Week 24
Remote Monitoring Frequency: Intermittent safety follow-ups as required by protocol
Checklist Review Interval: Regular caregiver reporting on participant's condition and medication compliance

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				